

6-Input and Output

Introduction

- Have you ever wondered how information gets into your computer and comes out in a form you can use?
 - Input devices convert what we understand into what the system unit can process
 - Output devices convert what the system unit has processed into a form that we can understand

What is Input?

- Any data or instructions entered into a computer
- Input devices translate data into a form that the system unit can process
- Some hardware input devices include:
 - Keyboards
 - Mice
 - Pointing
 - Scanning
 - Image capturing
 - Audio-input

Keyboard Entry

- Keyboards

- Traditional keyboards



- Laptop keyboards



- Virtual keyboard



- Thumb keyboard



Pointing Devices

Provide an intuitive interface by accepting pointing gestures and converting them into machine-readable input

- Wide variety of devices such as:

- Mouse
 - Touch screen
 - Game controller
 - Stylus

Mouse Types

- Optical mouse
 - Has no moving parts
 - Emits and senses light to detect mouse movement
 - Can be used on any surface
- Wireless mouse
 - Battery operated
 - Uses radio waves or infrared light waves
- Touch pads
 - Controls pointer by moving and tapping your fingers on the surface of the pad



Touch Screen

- Can be touched with more than one finger



- Stylus is a pen-like device
 - Used on tablets
 - Uses handwriting recognition software



Gaming Controllers

Provide input to computer games

- Joysticks use pressure and direction of the stick



- Gaming mice are similar to a mouse but high precision



- Game pads use both hands

- Motion sensing device control games by user movement



Scanning Devices

Scanners convert scanned data into a form the system unit can process

- Optical scanners
 - Flatbed scanners
 - Document scanners
 - Portable scanners
 - 3D scanners



Card Readers

Interpret encoded information that is stored on debit, credit and identification cards

- Magnetic card reader
 - Information read from strip when swiped through reader
- Chip card reader
 - Information read from a chip when inserted in reader
 - Smart cards hold additional security information

Bar Code Readers

Contain photo-electric cells that scan or read bar codes or the zebra striped marks printed on product containers

- Wand readers
 - Hand –held readers
- UPCs and MaxiCode readers
 - UPC are heavily used in grocery stores for automated checkout and inventory control
 - MaxiCode used by shipping companies for routing packages



RFID Readers



Radio-frequency identification

Tiny chips embedded in most anything contain electronically stored information that can be read using an **RFID reader** located several yards away.

- Tracking pets
- Update and control inventories
- Read passports

Character & Mark Recognition Readers



Recognize special characters and marks

- Character and mark recognition devices
 - Magnetic-ink character recognition (MICR)
 - Used by banks to read encoded characters on checks
 - Optical-character recognition (OCR)
 - Reads preprinted characters such as wand scanners
 - Optical-mark recognition (OMR)
 - Sense the presence of absence of marks used for test scoring

Image Capturing Devices

Create or capture original images

- Digital Camera
 - Capture images digitally and store in memory
- Webcams
 - Capture images and send to a computer

Audio-Input Devices

- Voice recognition systems
 - Use a microphone, sound card, and special software
 - Users can operate computers and create documents using voice commands
 - Included in many smart phones
 - Siri in iPhones
 - Cortana in Windows phones
 - Google Now in Google phones

Output



Processed data or information

- Types of output
 - Text
 - Graphics/photos
 - Audio & video
- Output devices
 - Monitors
 - Printers
 - Audio-output devices

Monitors



Known as screens or display screens and present visual images of text and graphics

- Features
 - Clarity
 - Resolution/pixels
 - Dot pitch
 - Contrast ratios
 - Size
 - Aspect ratio

Monitor Types

- Flat-panel monitors
 - Require less power to operate
 - Portable and thin
 - Most are backlit

Three types:

- Liquid Crystal Display (LCD)
 - Older monitors
- Light Emitting Diode (LED)
 - More advanced backlighting
- Organic Light Emitting Diode (OLED)
 - Thin layer organic compound that produces light

E-book Readers

An e-book is a traditional books printed in electronic form

E-book readers are dedicated mobile devices for storing and displaying e-books

- Use e-ink technology
 - Produce images that reflect light
 - Kindle
 - Nook

Other Monitor Types

- Other monitors
 - Digital/interactive whiteboards
 - Connects to a computer or project
 - Controlled using a special pen or even your finger
 - Classrooms and corporate boardrooms
 - Ultra High-definition television (UHDTV)
 - Digital output delivering a much clearer and more detailed image than regular HDTV
 - Digital Projector
 - Project the images from a traditional monitor onto a screen or wall

Printers

- Translates information that has been processed by the system unit
- Features
 - Resolution
 - Color
 - Speed
 - Memory
 - Duplex printing

Printer Types

- Ink-jet printers spray ink at a high speed
 - Reliable, quite and inexpensive
- Laser printers uses a laser light beam to produce images
 - Fast, excellent quality
 - Personal or shared
- 3D Printers create 3-D shapes with a thin layer of material repeatedly until created
 - Additive manufacturing

Other Printers

- Cloud printers
 - Connected to the Internet to provide services to others on the Internet
 - Thermal printers
 - Plotters

Audio and Video Devices

- Translates audio information from the computer into sounds that people can understand
 - Speakers and headphones
- Bluetooth Technology
 - Wireless technology
 - Used to connect to speakers and headsets

Combination Input and Output Devices

- Headsets
 - Combine a microphone and headphones
- Multifunctional devices (MFD)
 - Cost efficient but lower quality
 - All-in-one printers are a good example
- Virtual Reality (VR)
 - Artificial or simulated reality
- Virtual head-mounted displays and controllers

Drones

Drones or unarmed aerial vehicles

- Take input from a controller and the output device is the drone
- Very cost effective now

Robots

- Use cameras, microphones, and other sensors as inputs

Making IT Work for You ~ Head-Mounted Displays

- Two types of head-mounted displays
 - Virtual Reality (VR)
 - Augmented Reality (AR)
 - Can be expensive

Ergonomics

- Study of human factors related to things people use
- Fit the task to the user to avoid:
 - Eyestrain and headache
 - Back and neck pain
 - Repetitive strain injury



Ergonomic Challenged Devices

Portable devices are not set up for ergonomics

- Laptops
 - Because the keyboard and monitor are connected, they cannot be set up ergonomically
- Tablets
 - Tablet hunch is caused by the users head being improperly aligned to the viewing surface
- Smartphones
 - BlackBerry thumb results from using thumbs to type on a tiny keyboard

Careers In IT

- Technical writers prepare instruction manuals, technical reports, and other scientific or technical documents
- Typically requires an associates or bachelors degree in:
 - Communications
 - Journalism or
 - English
 - Specialization or familiarization with a technical field
- Technical writers can expect to earn \$54,000 to \$71,000 annually

A Look to the Future Internet of Things

- Smartwatch
 - Can monitor and share your location
 - Can count and share the number of steps
- Web-connected car
 - Track and share driving habits